

BelowTrace Detroit

Detroit is spending \$184M repairing sewer connections and does not publish where. We reverse engineered the City’s selection model from its own open data, so any household can see whether that money is coming to them.

IMPACT PILLAR

Reliable Transportation, Infrastructure & Sustainability

The repair money is already appropriated. What is missing is any way for the household on top of it to see where it goes.

TEAM

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Solo build. Research, data, design, engineering.

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A Detroit homeowner with sewage in the basement cannot find out whether the City is coming.



Why it is still unsolved. The \$18,400 alley program picks addresses itself and takes no applications. The \$30,000 repair grant requires proof of flood damage from June 2021, five years ago, and cannot pay for work already started. They sit in different departments, and the City's own guide states two different income limits on pages 3 and 10.

Photo: u/MarcRocket, r/Detroit, November 2025
BELOWTRACE DETROIT

1 in 3

Detroit homes have a failing private sewer connection, by DWSD's own director.

336

water in basement reports to Detroit 311 every month, for 42 months straight.

68,000

households no City program will reach, against roughly 11,500 it can.

“I go to homes with children’s bedrooms in one corner of the basement and feces in the floor in the opposite corner. I’ve called the city, I’ve emailed and never got a reply.”
A foundation repair contractor asking r/Detroit for any program he could refer families to. 184 upvotes, 55 replies, no working answer.

Rebuild the City's selection model from its own open data, and score every address against it.

1 Score the address

DWSD publishes four selection criteria and no list. We measure all four at one address: alley cave ins, block group income, council district, and distance to contracted work.

2 Validate against reality

Scored against the 138 alleys DWSD has actually contracted. Median basement flooding reports around them: 18, against 45 around the City's earlier alley projects. Permutation test $p < 0.0001$.

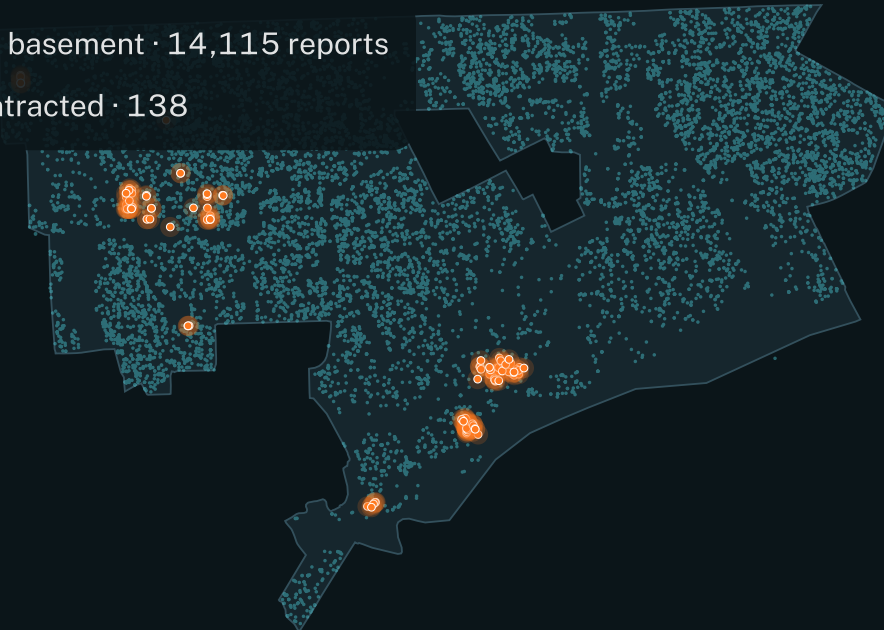
3 Then carry the call

There is no list to look up, so every answer ends in a phone call. Each number carries what to say, the three questions to ask, and what to write down before hanging up.

Why anyone comes back

A backup returns with every storm, the program rolls out in phases nobody publishes, and grant rounds open and close. The same household checks again after the next storm, before signing a contract, and before a sale. Each check is a decision worth \$15,000 to \$30,000.

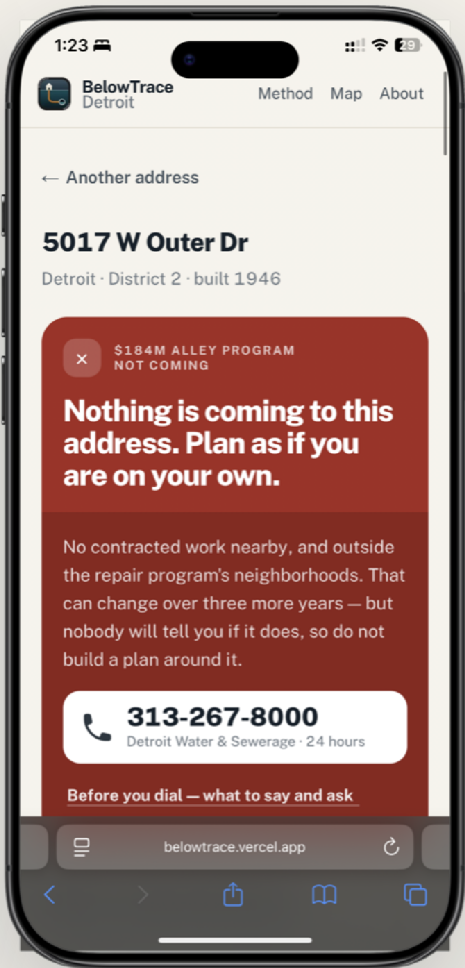
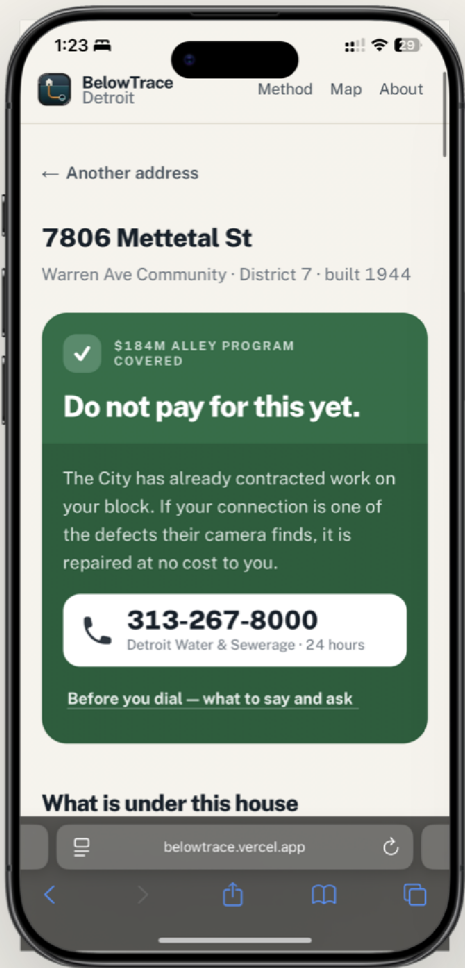
● Water in a basement · 14,115 reports
● Alleys contracted · 138



What the model found: the first round is not landing where basements flood.

14,115 flooding reports since 2023 against the 138 alleys contracted so far. 1.0% of the reports fall within 200 m of one, and four of seven council districts have none.

Two Detroit addresses, three miles apart. Opposite answers, in one tap.



✓

7806 METTETAL ST · DISTRICT 7

Do not pay for this yet.

Contracted work 213 m away. If their camera finds your connection, it is repaired at no cost to you.

✗

5017 W OUTER DR · DISTRICT 2

Nothing is coming to this address.

No alley in this district is in the first round. Plan as if you are on your own.

What every address already returns

✓

Six pipe segments, from the floor drain to the regional system, with the owner of each

✓

The alley program reading, with all five signals and the one that is not public

✓

Every City program checked, with the sequencing trap called out before you sign

✓

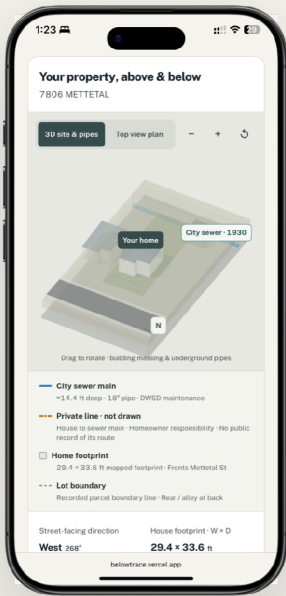
Scripted phone calls: what to say, three questions, what to write down

✓

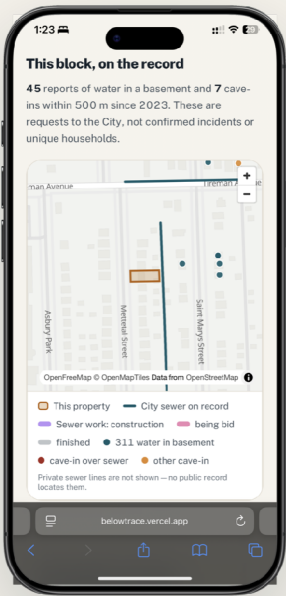
Permit history for the private line, and the 45 day claim letter for both agencies

Live at belowtrace.vercel.app. Free, no sign up, and the address report works on a phone with bad signal.

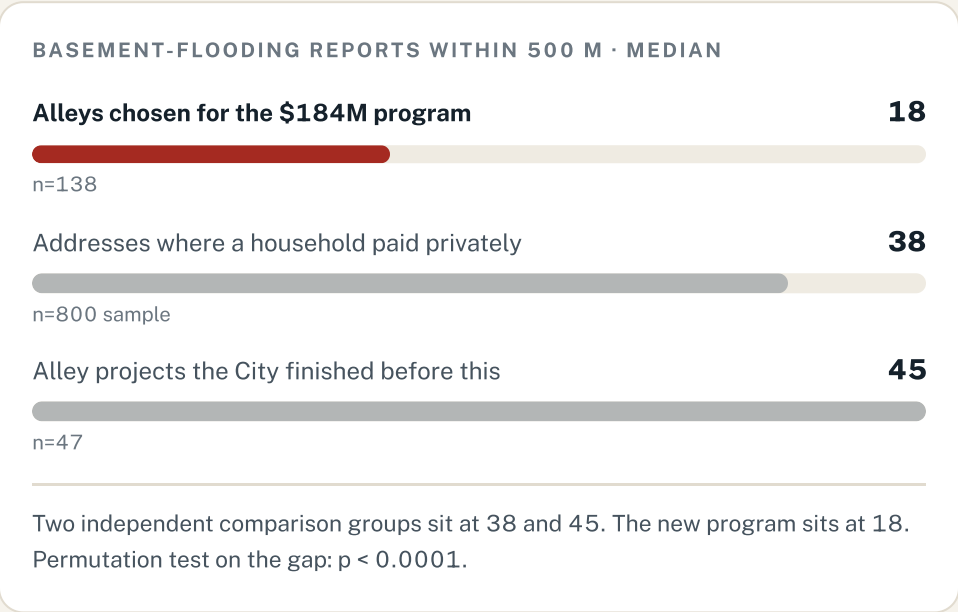
The reading is measured against the alleys the City actually chose.



What is under the house. The private line is never drawn.



Every 311 report and sewer project on the block.



We checked every ArcGIS service the City publishes, all 782. The 30,000+ failed connection points behind the selection are in none of them.

Roadmap

- NEXT** A verified list of contractors who accept state emergency relief. Michigan publishes none, so every household finds theirs by calling around.
- THEN** Spanish and Arabic. The City already publishes these programs in both.
- LATER** An alert when a contract reaches your alley, and a multi case view for intake staff.

The rough edges, named rather than hidden

First round of a four year program: 138 alleys against roughly 9,000 connections. DWSD’s camera evidence is not public, and the report says so on every address. 311 records are requests, not confirmed incidents.

Both groups lose money today. Neither one can tell which group it is in.

11,500

households the City can reach over four years

They stop paying for a repair already contracted. Detroit budgets about \$20,491 per connection, and a household that pays privately the month before the crew arrives never gets that back.

68,000

households no program will reach

They stop waiting for a notification that is never sent. Every year adds sewage exposure, mold, and lost property value to a repair that only gets more expensive.

And a public check on \$184M

Three years and most of the budget are still to be allocated, so publishing this now is the only moment it can change anything. Districts 2, 3, 4 and 5 have no alley in the first round.

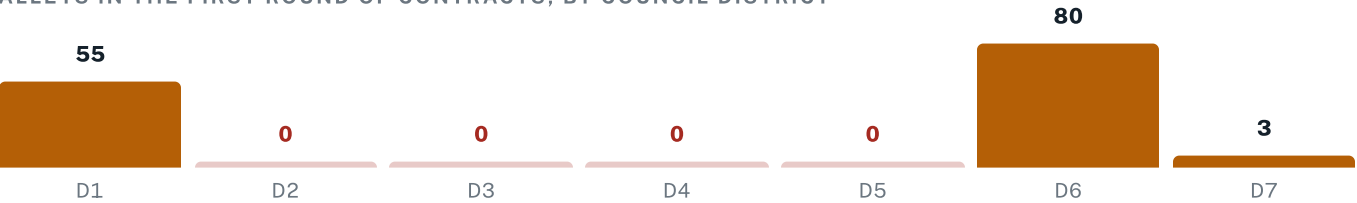
BEYOND DETROIT

The method travels. The program rules are configuration.

Homeowners own their sewer laterals in most American cities, and the three datasets this runs on exist almost everywhere: 311 requests, capital project phases, and census income.

What another city changes is the list of programs and the criteria of whichever agency holds the money. What it does not change is the approach: take the criteria an agency publishes, measure them against the work it has contracted, and give a household the answer.

ALLEYS IN THE FIRST ROUND OF CONTRACTS, BY COUNCIL DISTRICT



The City's CDBG-DR Action Plan named Districts 4, 6 and 7 as priorities. District 4 has none.

The first user is not the homeowner. It is whoever is standing in the basement with them.

FIRST USERS

The people who reach these households weekly and already ask this question for them: plumbers and waterproofing crews on service calls, community organizations doing repair intake, and council district staff. One contractor touches twenty households to find one who can pay. We reach them where they already ask.

SIZE

About one in three Detroit homes has a failing sewer connection, by DWSD's director. Detroit 311 logs 336 water in basement reports a month, and has for 42 months. 2,722 private sewer permits were pulled between 2019 and 2026, of which 1,288 were backwater valves bought by households protecting themselves.

SUSTAINABILITY

Every input is public open data and the app is static, with no AI at runtime and no accounts, so it costs close to nothing to run. The City's own repair policy already pays community organizations to do outreach and intake, which is the buyer shaped hole a screening tool fits. Nobody has agreed to buy it yet, and we say so.

The revenue we refuse

The obvious model is selling leads to contractors. We will not. The most valuable sentence this product says is “do not pay for this yet”, and it is worth nothing the moment anyone is paid when you do.

Early signal, stated honestly

What is real: the measurement reproduces from one script, and it found something no published City document shows. What is not: no resident has been observed using it to decide, and no agency has agreed to receive it. Those are the next things to find out, not things to claim.

THE ASK

One community partner running repair intake, and thirty minutes with DWSD.

The partner tells us whether this saves an intake worker time. DWSD tells us whether the reading is right, and whether they would publish the alley list themselves. Either answer improves the tool.

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